

CSE344 System Programming Homework 5 Report

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1 Introduction

This report presents the implementation of a cargo delivery simulator for CSE344 System Programming Homework 5. The simulator uses POSIX threads to manage multiple courier threads that process delivery orders from a priority queue. The system supports order prioritization, atomic statistics tracking, and safe shutdown via SIGINT signal handling.

2 Problem Description

The homework requires implementing a multi-threaded cargo delivery system with the following features:

- Multiple courier threads processing orders from a shared priority queue
- Priority-based ordering (EXPRESS > STANDARD > ECONOMY, then by order ID)
- Atomic statistics tracking for completed/cancelled orders and delivery times
- SIGINT-safe shutdown with proper cancellation of pending orders
- Required logging format for all operations
- Memory safety and thread synchronization

3 Design and Architecture

3.1 Thread Architecture

The system consists of the following threads:

- **Main Thread:** Initializes resources, loads orders, creates courier threads, waits for completion, and handles final statistics
- **Courier Threads:** Process orders from the priority queue, simulate deliveries, and update statistics
- **Signal Thread:** Dedicated thread for handling SIGINT signals safely

3.2 Data Structures

3.2.1 Priority Queue

A dynamic array-based priority queue that maintains sorted order by priority and ID:

```
1 typedef struct {
2     Order *orders;
3     uint32_t count;
4     uint32_t capacity;
5 } PriorityQueue;
```

3.2.2 Order Structure

Represents a delivery order with ID, recipient name, priority, and duration:

```
1 typedef struct {
2     uint32_t id;
3     char name[33];
4     Priority priority;
5     uint32_t duration;
6 } Order;
```

3.2.3 Global Statistics

Atomic counters for thread-safe statistics tracking:

```
1 typedef struct {
2     _Atomic(uint32_t) completed_orders;
3     _Atomic(uint32_t) cancelled_orders;
4     _Atomic(uint64_t) total_delivery_time;
5     CourierStats *couriers;
6     uint32_t num_couriers;
7     uint32_t total_orders;
8 } GlobalStats;
```

3.3 Synchronization

The system uses the following synchronization primitives:

- **Mutexes:** Protect shared data structures (queue, log output)
- **Condition Variables:** Coordinate between producer/consumer threads
- **Atomic Variables:** Thread-safe counters without locks

4 Implementation Details

4.1 Order Parsing

The `order_parse_line` function validates and parses input lines with strict format checking:

- ID: positive integer

- Name: ASCII alphanumeric + underscore, max 32 characters
- Priority: EXPRESS/STANDARD/ECONOMY
- Duration: positive integer

4.2 Priority Queue Operations

The queue maintains sorted order through insertion sort during enqueue operations:

```

1 int pq_insert(PriorityQueue *pq, const Order *order) {
2     // Find insertion position
3     uint32_t pos = pq->count;
4     for (uint32_t i = 0; i < pq->count; i++) {
5         if (order_compare(order, &pq->orders[i]) < 0) {
6             pos = i;
7             break;
8         }
9     }
10    // Shift and insert
11    // ...
12 }

```

4.3 Thread Management

4.3.1 Courier Thread Logic

Each courier thread follows this loop:

1. Wait for orders in queue (protected by mutex + condition)
2. Dequeue highest priority order
3. Simulate delivery using `nanosleep`
4. Update statistics atomically
5. Repeat until shutdown

4.3.2 Signal Handling

A dedicated signal thread uses `sigtimedwait` to handle SIGINT:

```

1 static void *signal_thread_func(void *arg) {
2     sigset_t sigset;
3     sigemptyset(&sigset);
4     sigaddset(&sigset, SIGINT);
5
6     while (!atomic_load(&shutdown_flag)) {
7         struct timespec timeout = {.tv_sec = 0, .tv_nsec = 1000000000};
8         int signo = sigtimedwait(&sigset, NULL, &timeout);
9         if (signo == SIGINT) {
10             atomic_store(&shutdown_flag, 1);
11             cleanup_shutdown();
12             break;
13         }

```

```
14     }  
15     return NULL;  
16 }
```

4.4 Statistics and Logging

All operations are logged to stdout with required format. Statistics are maintained using C11 atomic operations for thread safety.

5 Testing and Validation

5.1 Official Test Scenarios

The implementation was tested against all required scenarios:

1. **Scenario 1:** 10 mixed orders with 3 couriers
2. **Scenario 2:** 10 economy orders with 10 couriers
3. **Scenario 3:** SIGINT cancellation with 20 mixed orders
4. **Scenario 4:** Memory safety validation with Valgrind
5. **Scenario 5:** 20 mixed orders with 10 couriers

5.2 Additional Stress Tests

- Priority queue ordering verification
- SIGINT handling with active deliveries
- Invalid input handling
- CLI argument validation
- Large-scale test (200 orders, 5 couriers)

5.3 Memory Safety

Valgrind analysis confirmed:

- No memory leaks
- No invalid memory access
- All heap blocks freed

6 Results

6.1 Performance Metrics

Test results show correct behavior across all scenarios:

Scenario	Orders	Completed	Cancelled
1	10	10	0
2	10	10	0
3	19	Variable	Variable
4	10	10	0
5	19	19	0

Table 1: Test Scenario Results

6.2 Correctness Verification

- ✓ Priority queue maintains correct ordering
- ✓ SIGINT properly cancels pending orders
- ✓ Active deliveries complete before shutdown
- ✓ Statistics accurately reflect operations
- ✓ Thread synchronization prevents race conditions
- ✓ Memory management is leak-free

7 Conclusion

The implementation successfully meets all requirements of CSE344 HW5:

- Multi-threaded architecture with proper synchronization
- Priority-based order processing
- Atomic statistics tracking
- Safe SIGINT handling
- Required logging format
- Memory safety and performance

The system demonstrates robust concurrent programming practices and handles edge cases appropriately.